Danmarks Tekniske Universitet

Project Work - Bachelor of Artificial Intelligence and Data 02466

Adversarialness of Adversarial Attacks

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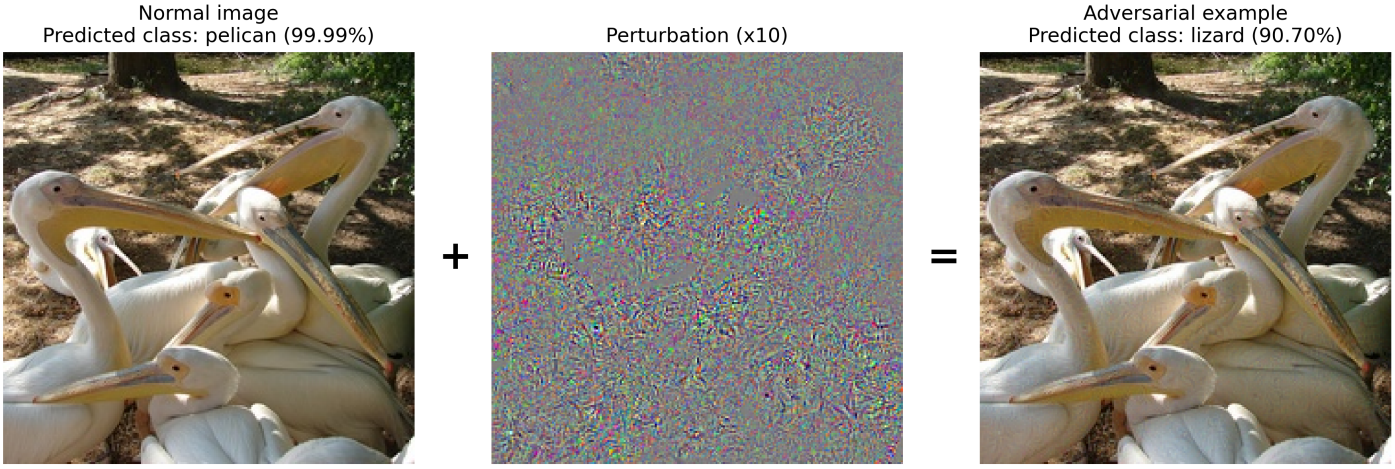


Figure 1: Predictions by ResNet-50 on a normal image and on an adversarial example (which is incorrectly classified as a "Lizard") created by the adversarial attack CW (with c = 20).

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1 Abstract

Adversarial attacks describe algorithms capable of degrading machine learning performance. Given this trait, it is crucial to quantify these attacks for further study. Past work has attempted to measure this but as of yet, there are no universally agreed-upon metrics within the literature. This project investigates the 'adversarialness' of adversarial attacks in machine learning by comparing the effectiveness, efficiency, and transferability of the three most popular white-box attack methods: Fast Gradient Sign Method (FGSM), Projected Gradient Descent (PGD), and Carlini & Wagner (CW). Using convolutional neural networks (ResNet-50 and VGG-16) pretrained on the ImageNet-100 dataset, we generate adversarial examples and evaluate their impact using a set of proposed metrics, including Robust Accuracy, True Class Robust Ratio, and transferability in terms of RA. Our results show that PGD is best at degrading model performance, CW produces the most imperceptible perturbations, and FGSM is the most computationally efficient and transferable across models. Furthermore, we find that ResNet-50 exhibits both greater clean accuracy and robustness than VGG-16 under all attacks, though neither model are capable of withstanding stronger perturbations, as their robust accuracies ends up plummetting to 0%. These findings emphasize the trade-offs between attack strength, subtlety, and generalizability while demonstrating the value of comprehensive evaluation metrics for understanding adversarial vulnerabilities in neural networks.

Indhold

1 Abstract **1**

2 Introduction **3**
2.1 Research Questions and Operationalization 3

3 Dataset **4**

4 Methodology **4**
4.1 Adversarial Attacks 4
4.1.1 Fast Gradient Sign Method (FGSM) 4
4.1.2 Projected Gradient Descent (PGD) 4
4.1.3 Carlini & Wagner (CW) 5
4.2 Evaluation Metrics 6
4.2.1 Robust Accuracy (RA) 6
4.2.2 True Class Robust Ratio (TCRR) 6
4.2.3 Attack Efficiency Metrics 7
4.3 Experimental Setup 8
4.3.1 Baseline Evaluation 8
4.3.2 Generating Adversarial Examples 8
4.3.3 Evaluating Attack Effectiveness 9
4.3.4 Transferability Evaluation 9
4.3.5 Reproducibility 9

5 Results **10**
5.1 Model Baseline Accuracy 10
5.2 FGSM 11
5.3 PGD 12
5.4 CW 13
5.5 Attack Efficiency Results 14
5.5.1 Average RA vs. average L_2-perturbation 14
5.5.2 Average RA vs. average SSIM-Drop 15
5.6 Transferability 16

6 Discussion **17**
6.1 Implication of Results 17
6.2 Societal and Ethical Reflections 19
6.3 Further Considerations and Scope Limitations 19

7 Conclusion **20**

8 Bibliography **21**

9 Appendix **22**
9.1 Robustness Ratio (RR) as a metric 22
9.2 Model finetuning epoch history 23

2 Introduction

From misclassified road signs in autonomous driving to anti-scraping tools like *Nightshade* [1], adversarial attacks have emerged as both a threat and a tool in the field of machine learning (ML). These attacks modify input data—often imperceptible to humans—to cause models to make incorrect predictions with high confidence. Such capabilities raise serious concerns about the security, robustness, and ethical use of ML systems.

At their core, adversarial attacks involve carefully crafted perturbations that can deceive models without being visual to human perception [2]. Their potential impact is particularly alarming in safety-critical applications. A subtle manipulation of a road sign, for example, could lead an autonomous vehicle to misinterpret a stop sign as a speed limit, an error with potentially life-threatening consequences. Simultaneously, adversarial techniques are being adapted for defensive use: *Nightshade* poisons images to prevent them from being used in unauthorized model training, thus empowering creators to resist large-scale data scraping.

Another important aspect of adversarial attacks is their transferability: the ability of adversarial examples generated for one model to successfully fool another, even when architectures differ. This means that attacks developed in white-box settings (where gradients are available) may still pose a threat to black-box models, for which the internals are unknown. Evaluating attack strength in both direct and transfer scenarios is essential for understanding real-world risk.

While many existing studies focus on defending against attacks or improving model robustness, less attention has been given to systematically comparing attack methods themselves. Not all adversarial attacks are equally effective or equally efficient: some cause significant performance drops with minimal perturbations, while others require more distortion to succeed.

This project investigates several approaches for quantifying adversarial impact, aiming to evaluate adversarial attack strength and efficiency of attacks using white-box methods as our testbed: FGSM, PGD, and CW. For this purpose, metrics such as Robust Accuracy (RA), True Class Robust Ratio (TCRR), and efficiency scores that relate accuracy degradation to perturbation magnitude were introduced. Attack transferability was also investigated across two pretrained convolutional neural networks: ResNet-50 and VGG-16. By systematically evaluating how adversarial attacks affect models under varying conditions, a contribution to a more nuanced understanding of what makes an adversarial attack effective, and what makes it genuinely adversarial, is made.

2.1 Research Questions and Operationalization

1. **How efficiently does each adversarial attack (FGSM, PGD, CW) reduce the accuracy of the model relative to the size of the perturbation?** The amount that each attack reduces the accuracy of the model is measured, normalized by the average magnitude of the perturbations introduced.
2. **How does each adversarial attack (FGSM, PGD, CW) impact the model's confidence in the true class prediction (True Class Robust Ratio)?** The proportion of the original confidence in the correct class that remains after an attack is measured, providing insight into how severely each method undermines the model's certainty, regardless of whether the final prediction changes.
3. **How transferable are adversarial examples crafted for a specific CNN to another CNN model, both trained on the same dataset?** Two CNNs (VGG16 and ResNet-50) are trained on the same dataset. Adversarial examples are then crafted on one source model using each adversarial attack (FGSM, PGD, CW) and test these examples on the target model to measure transferability via change in model accuracy.

3 Dataset

This study used ImageNet-100, a publicly available subset of ImageNet-1k, which contains 135,000 images from 100 randomly selected classes [3]. ImageNet-100 is chosen to ensure a balance between computational feasibility and dataset diversity, making it suitable for evaluating adversarial robustness in real-world settings. Furthermore, it contains only animal classes, which alleviates most of the ethical issues, concerning privacy, raised against the original ImageNet dataset [4].

The dataset statistics are as follows:

- Training Set: (80% of data) 108,000 images total, and 1,080 images per class.
- Validation Set: (10% of data) 13,500 images total, and 135 images per class.
- Test Set: (10% of data) 13,500 images total, 135 images per class.

The class splits are performed using stratified sampling, randomly selecting 80%, 10%, and 10% of the images within each class for the training, validation, and test sets, respectively. This ensures a balanced class representation across splits. A set seed of 42 was used to ensure reproducibility.

4 Methodology

4.1 Adversarial Attacks

This project implements three white-box adversarial attack methods: FGSM, PGD, and CW. All three attacks assume full access to model gradients and architecture. Each method perturbs image inputs in distinct ways, with different trade-offs in computational cost, perturbation size, and attack strength. In the following, the core mechanism behind each attack is described, along with the adaptations made for use in the experiments.

4.1.1 Fast Gradient Sign Method (FGSM)

FGSM is a rather simplistic adversarial attack that perturbs an input, an image in our case, in the direction of the gradient of the model's loss function [5]. Given a model with a loss function $J(\theta, x, y)$, where x is the input, x' is the perturbed input, y is the true label, and θ are the model parameters, FGSM generates an adversarial example as:

$$x' = x + \epsilon \cdot \text{sign}\left(\nabla_x J(\theta, x, y)\right) \tag{1}$$

Here, ϵ controls the perturbation magnitude, and $\text{sign}(\cdot)$ extracts the direction of the gradient that increases the loss with respect to the input.

As the formula shows, the attack is relatively simple and computationally efficient. The primary concept is that FGSM takes a step in the direction that maximizes the loss function for the given input. The idea is to choose an ϵ that is small but often sufficient to mislead the classifier. Since FGSM only requires a single gradient computation, it is an extremely fast algorithm with a time complexity of $\mathcal{O}(1)$.

4.1.2 Projected Gradient Descent (PGD)

PGD extends FGSM by iteratively refining the perturbation while keeping it within a bounded region to ensure more effective misclassification [6]:

$$x_{t+1} = \text{clip}_{x,\epsilon}\big(x_t + \alpha \cdot \text{sign}(\nabla_x J(x_t, y))\big) \tag{2}$$

Where α defines the step size per iteration. The introduced *clipping function* is defined as:

$$\text{clip}_{x,\epsilon}(x') = \min(\max(x', x - \epsilon), x + \epsilon)$$

Here, x denotes the original, unperturbed input, while x' represents the adversarial example. The parameter ϵ once again controls the maximum allowable perturbation magnitude. Intuitively, the clipping function acts as a guardrail that ensures the adversarial example stays within a permitted range of distortion, ϵ. Specifically, it projects the perturbed input back into an ϵ-bounded region around the original input after each step. Compared to FGSM, PGD considered to be more severe due to its iterative nature. However, these very same iterations also make it more computationally demanding than FGSM, demanding a runtime of $\mathcal{O}(T)$, where T denotes the number of iterations.

4.1.3 Carlini & Wagner (CW)

CW is an optimization-based attack that aims to minimize a carefully designed loss function while keeping perturbations imperceptible [7]. The attack solves the following optimization problem:

$$\min_{x'} \left(D(x, x') + c \cdot f(x')\right) \tag{3}$$

Here, $D(x, x')$ is a distance metric that quantifies how much the perturbed input x' deviates from the original input x. CW originally explores the use of the L_0, L_2, and L_∞ norms, which correspond to the number of pixels perturbed, the Euclidean distance between the inputs, and the maximum change to any individual pixel, respectively. The constant c controls the trade-off between minimizing the perturbation size and ensuring the attack is successful.

The loss function $f(x')$ serves as the adversarial loss that encourages misclassification. It is typically defined as:

$$f(x') = \max(\max_{i \neq t}\{Z(x')_i\} - Z(x')_t, -\kappa)$$

where $Z(x')$ denotes the model's logits (the raw class scores before softmax) and t is the target class in a targeted attack (or the original class in an untargeted one). The parameter κ acts as a confidence threshold: larger values of κ push the optimizer to find adversarial examples that are not just misclassified, but misclassified with a greater margin. If the adversarial example already achieves the desired misclassification, increasing κ prevents the optimizer from introducing unnecessary perturbations.

Due to the c constant balancing perturbation size and misclassification, CW is capable of generating imperceptible adversarial examples by solving its constrained optimization problem. Unlike FGSM, which perturbs in a fixed direction, CW iteratively searches for minimal perturbations that induce misclassification, rendering it highly effective. However, due to its iterative nature, the algorithm is also the most computationally expensive of the three attacks. CW has a time complexity of $\mathcal{O}(N \cdot I)$ where N and I denotes the amount of binary search steps and number of optimization iterations per search, respectively.

4.2 Evaluation Metrics

4.2.1 Robust Accuracy (RA)

Robust Accuracy (RA) is defined as the proportion of samples that were correctly classified by the model on clean inputs that remain correctly classified after an adversarial attack. It measures how well a model maintains correct predictions under perturbation.

$$\boxed{\text{RA} \triangleq \frac{1}{N_{\text{correct}}} \sum_{i \in \mathcal{I}_{\text{correct}}} \mathbf{1}\left[\hat{y}_i^{\text{adv}} = y_i\right]} \tag{4}$$

Let y_i denote the true label for image/sample i. The model's prediction on the clean input x_i is given by $\hat{y}_i^{\text{clean}}$, while $\hat{y}_i^{\text{adv}}$ denotes the prediction on the adversarially perturbed input x_i^{adv}. We define the set of indices where the model's prediction on clean inputs is correct as $\mathcal{I}_{\text{correct}} = \{i \mid \hat{y}_i^{\text{clean}} = y_i\}$, and let $N_{\text{correct}} = |\mathcal{I}_{\text{correct}}|$ denote the number of such samples. The indicator function $\mathbf{1}[\hat{y}_i = y_i]$ returns 1 if the model's prediction equals the true label and 0 otherwise.

4.2.2 True Class Robust Ratio (TCRR)

While RA measures whether a model's predictions remain correct after an adversarial attack, it provides only a binary outcome and fails to capture how the attack affects the model's confidence in the correct class. To gain a deeper understanding of robustness, we introduce the *True Class Robust Ratio (TCRR)*, a metric that quantifies how adversarial perturbations change the model's predicted probability for the ground truth class. This allows us to assess how much an attack undermines the model's belief in the true label, even when it does not cause a misclassification.

Let:

- P_{clean} be the model's predicted probability for the true class y_i on the clean input x_i,
- P_{adv} be the predicted probability for y_i on the adversarial input x_i^{adv}.

We initially considered defining TCRR as the drop in confidence of the true class:

$$1 - (P_{\text{clean}} - P_{\text{adv}})$$

However, this formulation fails to account for the significance of the drop relative to the model's initial certainty. For example, a drop from 0.9 to 0.6 is treated the same as a drop from 0.3 to 0.0, despite the latter being more severe in relative terms. To better represent this difference, we instead divide the confidence drop by the original (clean) confidence:

$$1 - \frac{P_{\text{clean}} - P_{\text{adv}}}{P_{\text{clean}}}$$

Which conveniently simplifies to:

$$\frac{P_{\text{adv}}}{P_{\text{clean}}}$$

This ratio reflects how much of the model's original confidence in the true class is retained after an adversarial perturbation for a single image. A value close to 1 means the attack had little impact on the model's trust in the correct label, while a value near 0 indicates a significant degradation.

To compute the TCRR across an entire dataset of size N, we take the mean:

$$\boxed{\text{TCRR} \triangleq \frac{1}{N} \sum_{i=1}^{N} \frac{P_{\text{true},i}^{\text{adv}}}{P_{\text{true},i}^{\text{clean}}}} \tag{5}$$

TCRR therefore provides a continuous, interpretable measure of how adversarial attacks affect the model's confidence in the ground truth, revealing whether an attack can significantly reduce that confidence under any circumstance — even if the final classification remains unchanged. This complements RA by highlighting subtler effects of adversarial perturbations that may not lead to misclassification but still undermine the model's reliability[1].

4.2.3 Attack Efficiency Metrics

While RA quantifies the impact of adversarial attacks in terms of accuracy loss, it does not account for how much distortion/perturbation was required to cause that degradation. In practice, increasing parameters like ϵ (for FGSM and PGD) or c (for CW) will lead to stronger perturbations and thus a greater drop in accuracy, but this alone does not make an attack truly adversarial. What makes these examples concerning is their *imperceptibility*: the model fails even when the input appears virtually unchanged to the human eye.

In this light, two attacks achieving the same RA can differ dramatically in how much they needed to distort the input. To capture this distinction, we introduce an efficiency metric that normalizes the drop in RA by the average perturbation magnitude.

Perturbation Magnitude via Mean L_2 Norm We quantify perturbation magnitude using the average L_2 norm between clean and adversarial examples across all N test samples:

$$\text{Mean } L_2 = \frac{1}{N} \sum_{i=1}^{N} \left\| x_i^{\text{adv}} - x_i \right\|_2 \tag{6}$$

This provides a direct measure of how far, on average, the inputs were shifted in Euclidean space.

Efficiency Score The efficiency of an attack is then defined as the drop in Robust Accuracy per unit of perturbation, computed as:

$$\boxed{\text{Efficiency Score} = \frac{1 - \text{RA}}{\text{Mean } L_2}} \tag{7}$$

This metric allows us to compare attacks based on how effectively they degrade model accuracy at a given perturbation magnitude.

Perceptual Difference via SSIM Drop Since L_2 norms do not always align with human perception, we also compute the Structural Similarity Index Measure (SSIM) [8] between clean and adversarial images:

$$\text{SSIM}(x, x') = \frac{(2\mu_x\mu_{x'} + C_1)(2\sigma_{xx'} + C_2)}{(\mu_x^2 + \mu_{x'}^2 + C_1)(\sigma_x^2 + \sigma_{x'}^2 + C_2)} \tag{8}$$

[1] See Appendix∼9.1 for why we include TCRR over the standard Robust Ratio (RR)

Here, μ, σ, and $\sigma_{xx'}$ denote image statistics (means, variances, covariance), and C_1, C_2 are constants for numerical stability. SSIM ranges from -1 to 1, with 1 indicating perfect similarity.

We compute the average perceptual degradation as:

$$\text{Mean SSIM Drop} = \frac{1}{N} \sum_{i=1}^{N} \left(1 - \text{SSIM}(x_i, x_i^{\text{adv}})\right)$$

This is used to generate RA-vs-SSIM plots, which illustrate how each attack affects classification accuracy as inputs become more visibly distorted. A threshold at a 0.05 SSIM drop is used to mark the onset of human-perceptible changes [9], helping us identify which attacks are genuinely adversarial—i.e., successful without being noticeable.

4.3 Experimental Setup

To evaluate the effectiveness and transferability of adversarial attacks, we conduct a series of controlled experiments using two pretrained convolutional neural networks: ResNet-50 and VGG-16 [10, 11]. These models are selected due to their architectural diversity and widespread use in both standard and adversarial machine learning research. ResNet-50 is a deep residual network known for strong generalization, while VGG-16 offers a simpler, baseline architecture that allows us to investigate how adversarial vulnerability varies with network complexity.

All experiments are conducted on the test set of the ImageNet-100 dataset (see Section 3) with a fixed seed used for all randomness to ensure result reproducibility. We begin by evaluating both models on clean, unperturbed data to establish a performance baseline.

4.3.1 Baseline Evaluation

We begin by measuring the top-1 classification accuracy of both models on clean (unperturbed) test image split of the ImageNet-100 dataset. These baseline accuracies provide reference points to quantify the degradation caused by each adversarial attack.

4.3.2 Generating Adversarial Examples

AEs are generated using three white-box methods: FGSM, PGD, and the CW attack. Each attack perturbs the input image using model gradients to induce misclassification, while aiming to keep the changes imperceptible.

FGSM and PGD Configuration: For FGSM and PGD, we vary the perturbation strength ϵ from 0.01 to 0.25 in increments of 0.02. PGD is configured to run for 10 iterations, each with a step size set to $\alpha = \epsilon/10$. After every step, the perturbed input is projected back into the ϵ-bounded region surrounding the original image to enforce norm constraints and maintain bounded perturbation. These settings follow the recommendations of the original FGSM paper, which demonstrated optimal performance within this range [5].

CW Attack Configuration: For the CW attack, we adopt the L_2 norm as the perturbation metric and vary the optimization parameter c across a linear range of values: $\{0.1, 0.5, 1, 2, 5, 10, 20\}$. The confidence parameter κ is fixed at 0, focusing the optimization on minimally perceptible adversarial examples. Rather than using a binary search to fine-tune c as proposed by Carlini and Wagner [7], we opt for a simpler linear sweep to cover a broader and more interpretable spectrum of perturbation strengths. All CW-generated images are evaluated using both the resulting L_2 norm and SSIM score to assess distortion and visual similarity.

4.3.3 Evaluating Attack Effectiveness

To evaluate the impact of each attack, we compute both classification-based and distortion-aware metrics for each perturbation parameter value. RA, which measures the proportion of originally correct predictions that remain correct after perturbation, and the TCRR, which captures the relative reduction in the model's confidence in the correct class. In addition, we compute the RA-L_2 Efficiency Score, defined as the accuracy degradation normalized by the average L_2 perturbation magnitude, and the SSIM Drop, which quantifies perceptual differences between clean and adversarial images.

4.3.4 Transferability Evaluation

To study how well adversarial perturbations generalize across models, we generate AEs using a source model (e.g., ResNet-50) and evaluate them on a target model (e.g., VGG-16). This cross-model evaluation reveals whether perturbations crafted for one architecture transfer to another, a critical aspect of real-world adversarial robustness.

4.3.5 Reproducibility

All experiments, attack implementations, evaluation scripts, and model configurations are available in the project's GitHub repository [12] to ensure full transparency and reproducibility.

5 Results

5.1 Model Baseline Accuracy

Before generating adversarial examples, the baseline accuracy of the ResNet-50 and VGG-16 models were first measured on the unperturbed ImageNet-100 test set. This provided a reference point for evaluating the impact of each attack.

Model	**Mean Clean Accuracy**	**95% Confidence Interval**
VGG16	75.79%	[75.070%, 76.515%]
ResNet-50	92.90%	[92.470%, 93.337%]

Tabel 1: Mean standard accuracy and 95% confidence intervals for VGG16 and ResNet-50 on the unperturbed test dataset.

As shown in Table∼1 the confidence intervals of the model accuracies do not overlap: therefore ResNet-50 performed better than VGG16 at a stastically significant level.[2] What is interesting to note, is that despite the worse performance of the VGG16 model, it not only took longer to train but also required significantly more time to run adversarial attack evaluations on. When looking at pure mean clean accuracy, however, it unfortunately still performs significantly worse than ResNet-50. This difference in performance is likely explained by their different model architecture: VGG16 is a relatively simple model with uniform architecture whereas ResNet-50 is 'deeper' and generally capable of doing more complicated computer vision tasks.

After establishing these baselines, FGSM, PGD, and CW attacks were applied across 13,500 test images and analyzed their effects on accuracy and True Class Robust Ratio (TCRR) under varying perturbation magnitudes.

[2]see Appendix∼9.2 for details about model construction

5.2 FGSM

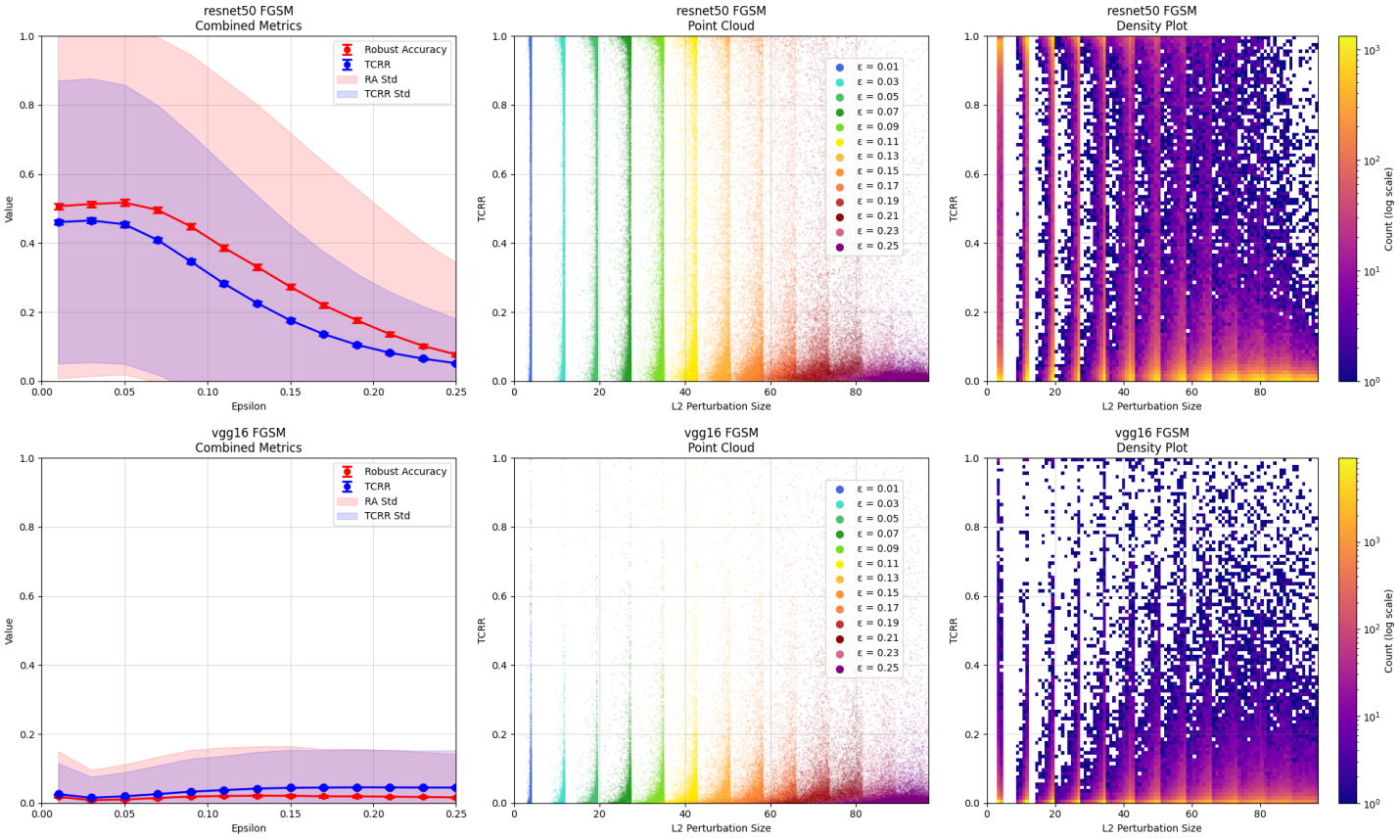


Figure 2: Performance of FGSM at different ϵ values on ResNet-50 and VGG-16. RA and TCRR are displayed with their 95% confidence intervals and SD. Notice that the CIs are narrow on the lefthand plots due to large sample size but that the data still contains high standard deviations.

For ResNet-50, FGSM shows limited model degradation at low ϵ values. The RA and TCRR initially go up slightly, before declining more consistently as ϵ increases (see Figure 2). The final RA approaches 0.1 with a standard deviation of approximately 0.35. TCRR plots against the L2 norm reveals scattered results, with many samples concentrated at both TCRR = 1 and TCRR = 0, a trend that continues for perturbations up to the equivalent of 60 full pixel-channel changes. Only beyond this point did TCRR values begin to cluster around 0–0.15, indicating a more consistent reduction in true-class confidence.

For VGG-16, FGSM was significantly more destructive. RA dropped to nearly 0 even at low ϵ values and remained that way, with a standard deviation around 0.15. TCRR values were consistently low across all perturbation magnitudes, showing a total collapse in TCRR.

5.3 PGD

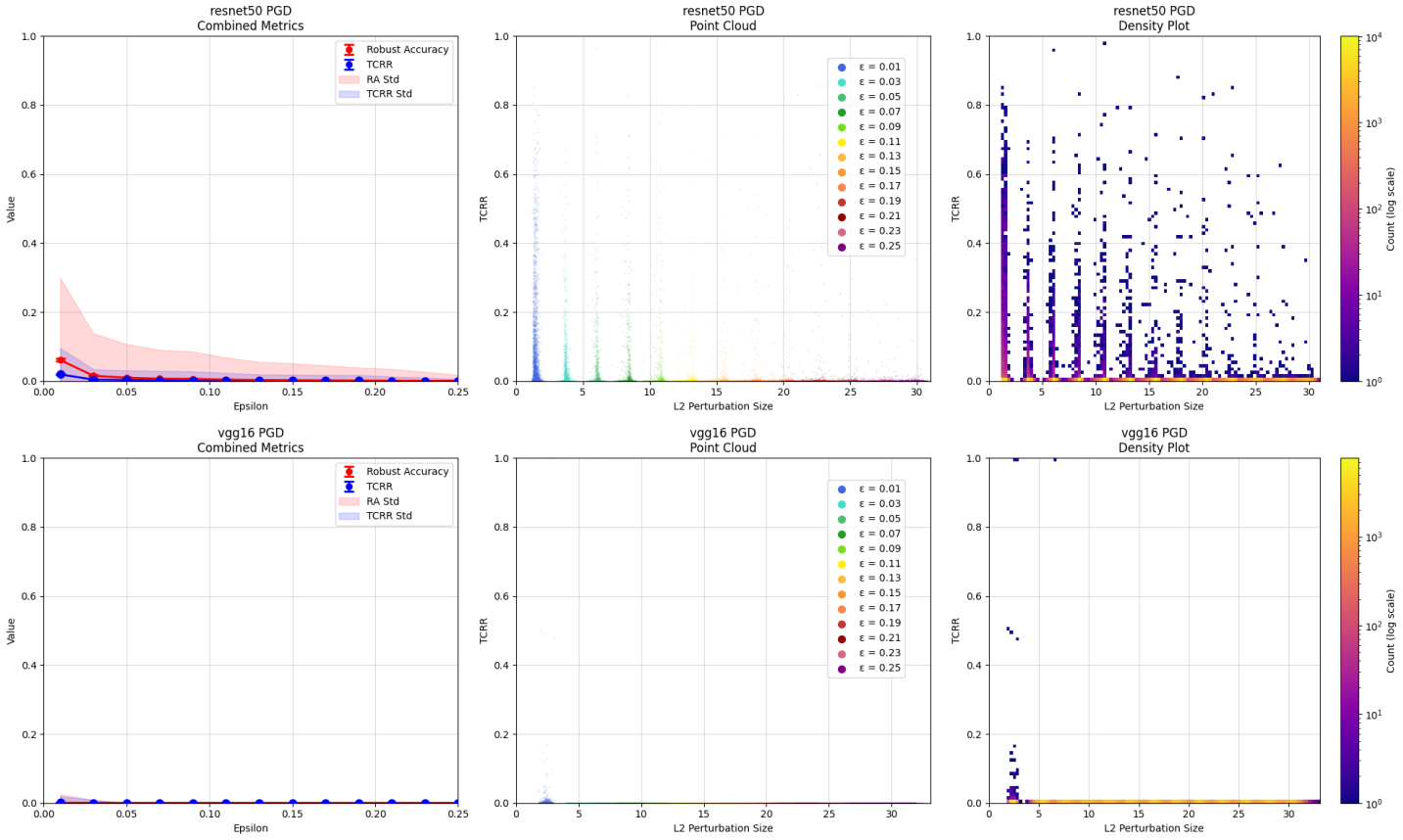


Figure 3: Performance of PGD at different ϵ values on ResNet-50 and VGG-16. RA and TCRR are displayed with their 95% confidence intervals and SD. Notice that the CI is narrow due to large sample size but that the data still contains high standard deviations, as shown.

When looking at model performance degradation in the lefthand plots on Figure 3, it is quite clear that PGD severely affects them both in regards to RA and TCRR. The ResNet-50 model RA rapidly drops at the first few ϵ values and remained at zero for all subsequent values, with a standard deviation of about 0.2 or less. TCRR also dropped rapidly, although a few scattered examples still retained higher values of TCRR.

On VGG-16, PGD is even more severe in its attack. Accuracy remains at zero from the start, and TCRR values are tightly clustered around 0, with minimal variance. Compared to ResNet-50, there are fewer examples that retains any measurable confidence in the true class, indicating that VGG-16 is less robust under this type of attack.

5.4 CW

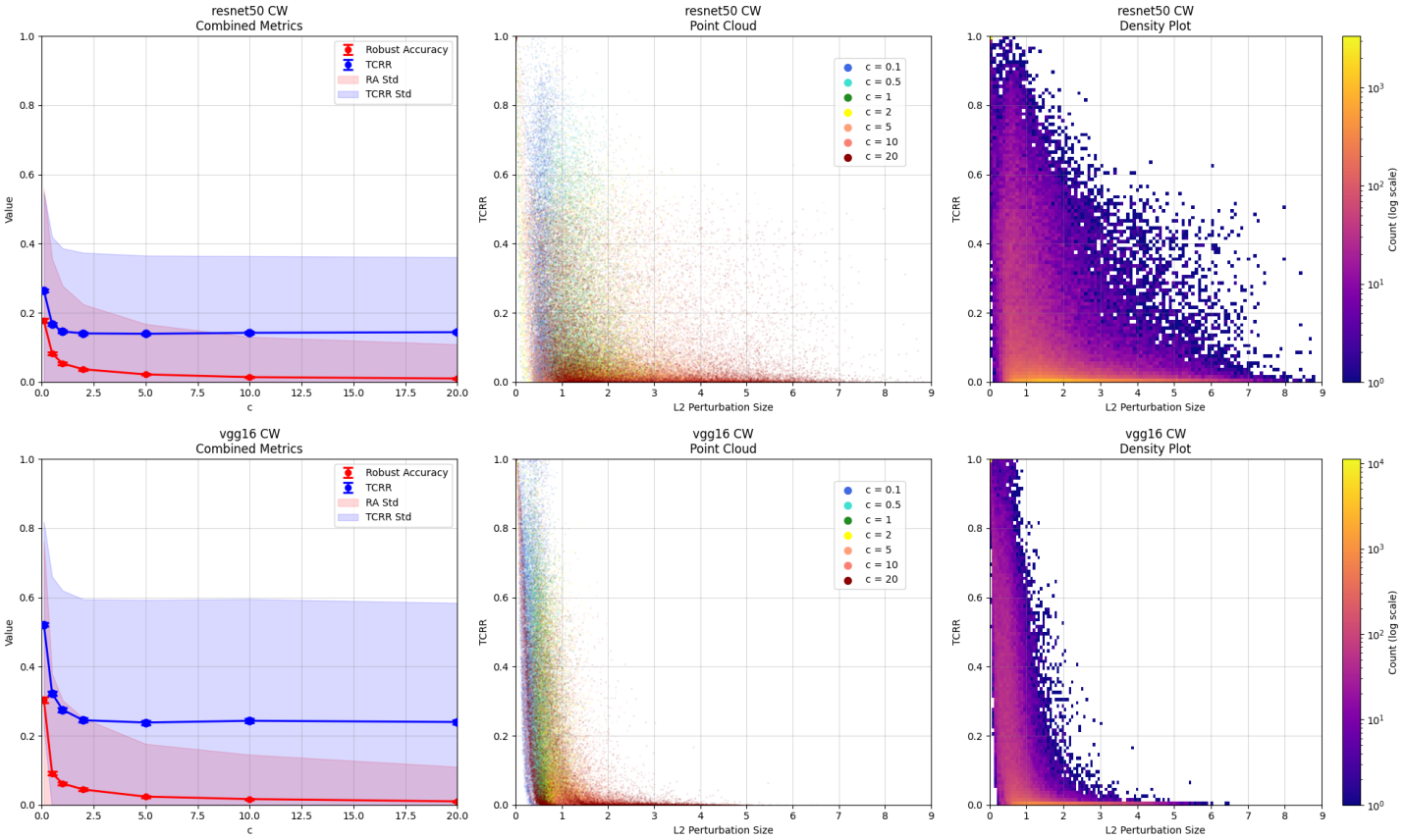


Figure 4: Performance of CW at different c values on ResNet-50 and VGG16. RA and TCRR are displayed with their 95% confidence intervals and SD. Notice that the CI is narrow due to large sample size but that the data still contains high standard deviations, as shown.

The CW attack also quick drop-off in regards to especially RA on both CNN models, although slightly below PGD. Perturbation-wise, however, it generats the smallest perturbations of all three attacks. On ResNet-50, the initial RA is approximately 0.2, and it dropped close to zero as the balancing parameter c increased. For VGG-16, the drop was even more dramatic, starting at roughly 0.3 and reaching close to 0 by $c \approx 5$–7.5.

TCRR also drops rapidly for both models. In many cases, just a single pixel-channel perturbation is enough to reduce the true-class confidence to nearly 0.

5.5 Attack Efficiency Results

5.5.1 Average RA vs. average L_2-perturbation

Tabel 2: Comparison of FGSM, PGD, and CW Efficiency Scores on ResNet-50 and VGG16.

Attack	**Param**	**ResNet-50**				**VGG16**			
		Rob. Acc	**Degrad.** *(RA - 1)*	**Avg** L_2	**Eff** L_2	**Rob. Acc**	**Degrad.** *(RA - 1)*	**Avg** L_2	**Eff** L_2
FGSM	0.01	0.5453	0.4547	3.8499	0.1181	0.0235	0.9765	3.8496	0.2537
	0.03	0.5519	0.4481	11.4930	0.0390	0.0104	0.9896	11.4917	0.0861
	0.05	0.5565	0.4435	19.0702	0.0233	0.0137	0.9863	19.0674	0.0517
	0.07	0.5325	0.4675	26.5790	0.0176	0.0190	0.9810	26.5740	0.0369
	0.09	0.4799	0.5201	34.0141	0.0153	0.0240	0.9760	34.0066	0.0287
	0.11	0.4129	0.5871	41.3708	0.0142	0.0255	0.9745	41.3605	0.0236
	0.13	0.3525	0.6475	48.6447	0.0133	0.0256	0.9744	48.6315	0.0200
	0.15	0.2910	0.7090	55.8319	0.0127	0.0254	0.9746	55.8155	0.0175
	0.17	0.2337	0.7663	62.9271	0.0122	0.0228	0.9772	62.9077	0.0155
	0.19	0.1861	0.8139	69.9252	0.0116	0.0218	0.9782	69.9031	0.0140
	0.21	0.1433	0.8567	76.8205	0.0112	0.0209	0.9791	76.7957	0.0127
	0.23	0.1072	0.8928	83.6073	0.0107	0.0195	0.9805	83.5804	0.0117
	0.25	0.0810	0.9190	90.2802	0.0102	0.0186	0.9814	90.2515	0.0109
PGD	0.01	0.0653	0.9347	1.4620	0.6393	0.0008	0.9992	2.4477	0.4082
	0.03	0.0164	0.9836	3.7838	0.2599	0.0001	0.9999	5.3896	0.1855
	0.05	0.0103	0.9897	6.1359	0.1613	0.0000	1.0000	7.7164	0.1296
	0.07	0.0074	0.9926	8.4955	0.1168	0.0000	1.0000	9.9454	0.1005
	0.09	0.0068	0.9932	10.8518	0.0915	0.0000	1.0000	12.1633	0.0822
	0.11	0.0044	0.9956	13.2029	0.0754	0.0000	1.0000	14.3799	0.0695
	0.13	0.0030	0.9970	15.5460	0.0641	0.0000	1.0000	16.6001	0.0602
	0.15	0.0026	0.9974	17.8805	0.0558	0.0000	1.0000	18.8257	0.0531
	0.17	0.0020	0.9980	20.2039	0.0494	0.0000	1.0000	21.0547	0.0475
	0.19	0.0015	0.9985	22.5194	0.0443	0.0000	1.0000	23.2856	0.0429
	0.21	0.0012	0.9988	24.8239	0.0402	0.0000	1.0000	25.5146	0.0392
	0.23	0.0007	0.9993	27.1198	0.0368	0.0000	1.0000	27.7422	0.0360
	0.25	0.0003	0.9997	29.4015	0.0340	0.0000	1.0000	29.9665	0.0334
CW	0.1	0.1907	0.8093	0.6699	1.2080	0.3991	0.6009	0.3114	1.9296
	0.5	0.0879	0.9121	1.2946	0.7045	0.1194	0.8806	0.5837	1.5086
	1.0	0.0572	0.9428	1.6171	0.5830	0.0812	0.9188	0.7794	1.1789
	2.0	0.0393	0.9607	1.9433	0.4944	0.0582	0.9418	1.0313	0.9132
	5.0	0.0234	0.9766	2.2873	0.4270	0.0313	0.9687	1.3496	0.7178
	10.0	0.0148	0.9852	2.4662	0.3995	0.0221	0.9779	1.5360	0.6367
	20.0	0.0106	0.9894	2.5722	0.3846	0.0133	0.9867	1.5814	0.6239

Table 2 displays some general trends for both models: as the perturbation magnitudes ϵ and c increase, the L_2 norm increases as well (as shown on the previous graphs), whereas Efficiency Score decreases correspondingly. This effect is especially pronounced for FGSM and PGD which shows that they tend to utilize most of the perturbation budget. This leads to greater model degradation but carries the risk of perceptible changes. Conversely, CW always attempts to create imperceptible adversarial examples when $\kappa = 0$.

5.5.2 Average RA vs. average SSIM-Drop

To measure the efficiency of adversarial attacks model degradation was compared to perturbation magnitude. This was done by constructing a graph of mean RA against the mean SSIM drop on both ResNet-50 and VGG16 performance.

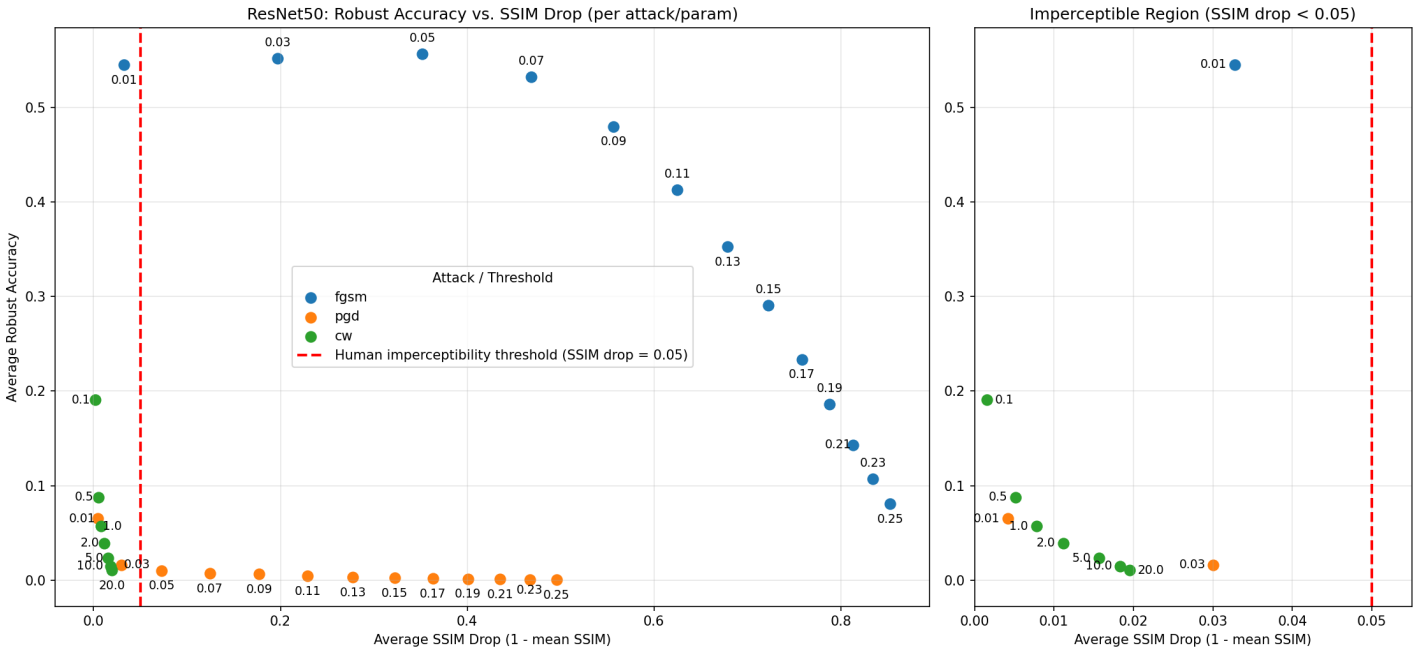


Figure 5: Left image: average robust accuracy plotted aginst mean SSIM-drop. Right image: zoomed in version of the imperceptible region.

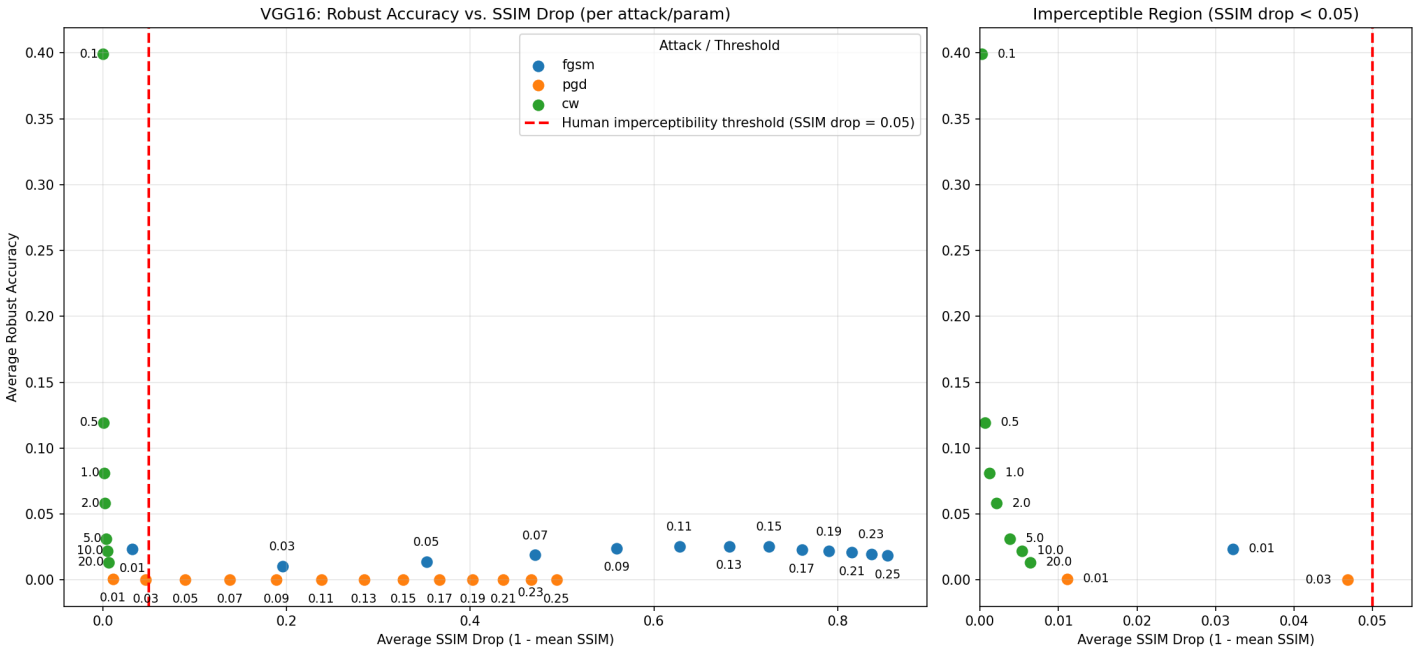


Figure 6: Left image: average robust accuracy plotted aginst mean SSIM-drop. Right image: zoomed in version of the imperceptible region.

When observing the results, it confirms inferred information from the L_2 plots in the previously shown results: most FGSM and PGD perturbations tend to exceed the human imperceptibility threshold - only the lowest and second lowest ϵ values manage to contain themselves within the threshold, respectively. Essentially, it means that FGSM and PGD produce perturbations visible to the human eye. This can be explained by the fact that both algorithms attempt to capitalize on the given perturbation magnitude, ϵ, and therefore attempt to aggressively exploit the perturbation 'budget'.

CW, on the other hand, is contained within the imperceptibility threshold for all the tested c values. Its adversarial examples are therefore imperceptible. It is therefore the most efficient adversarial attack. This behavior is expected since κ was set to 0, prompting the attack to minimize perturbations.

5.6 Transferability

In order to test the transferability of adversarial attacks, adversarial examples were first generated using FGSM, PGD, and CW on ResNet-50 as the source model before testing them on VGG16 as the target model and vice versa.

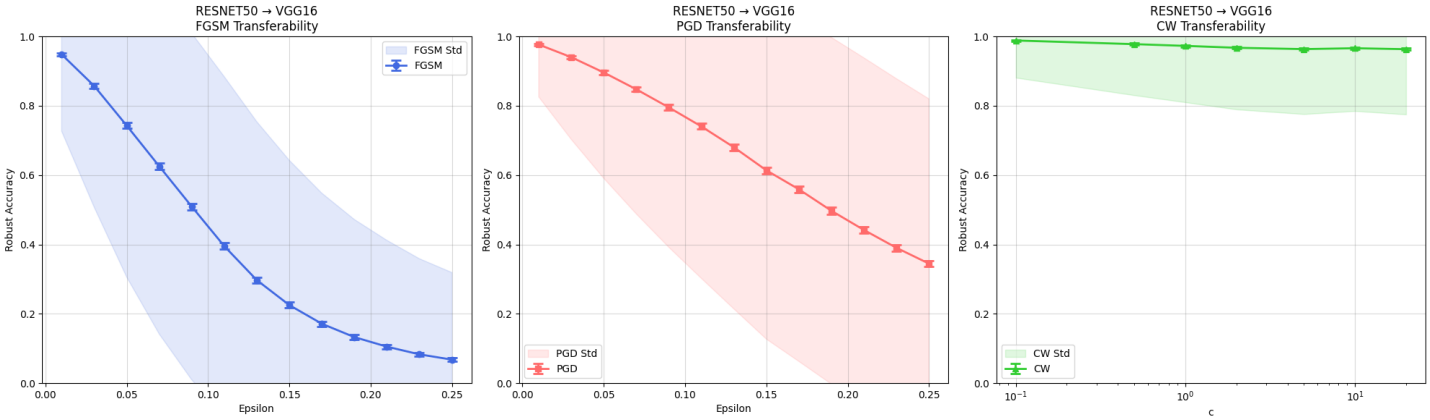


Figure 7: Transferability results with ResNet-50 as source model and VGG-16 as target model.

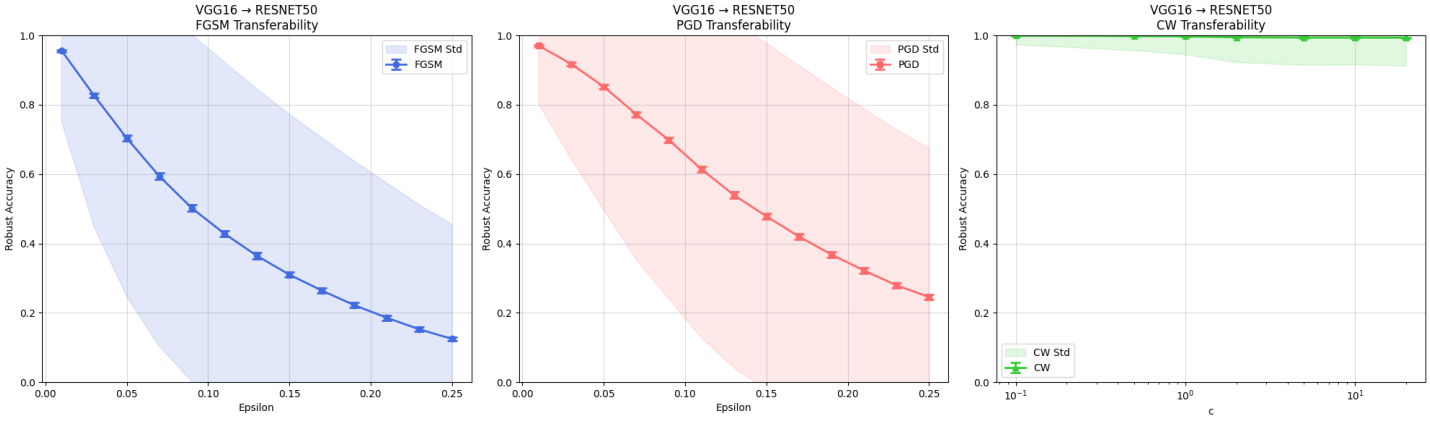


Figure 8: Transferability results with VGG-16 as source model and ResNet-50 as target model.

Looking at the results in Figure 7 and 8, both FGSM and PGD cause low RA at higher ϵ values, regardless of source and target models. This indicates that both attacks display a high degree of transferability. CW, on the other hand, does a very poor job at reducing RA when transferring from ResNet-50 to VGG16 and is entirely unable to lower it the other way around, showing extremely poor transferability capabilities. In general it seems like FGSM transfers best, followed by PGD and then CW.

6 Discussion

6.1 Implication of Results

Across all experiments, ResNet-50 consistently showed greater robustness compared to VGG-16, particularly in retaining classification accuracy and true-class confidence under increasing perturbation magnitudes. Among the three attacks, PGD (even with fewer iterations) emerged as the most powerful, followed by CW, with FGSM being the weakest and most unstable. This result reached statistical significance.

Notably, for both models, FGSM appeared to exploit the full perturbation budget more aggres-

sively at initial, lower ϵ-values, therefore perturbing the images more aggressively and creating these point-cloud "lines". As the ϵ-value was increased, it resulted in more evenly distributed perturbations — an interesting, though likely unintentional, artifact of the attack's one-shot nature. This aligns with the mathematical theory of the attacks and therefore lies within expectations.

CW followed PGD closely in model degradation but was also able to generate the least perceptible attacks. This confirms CW's precision in finding minimal but highly effective perturbations. A possible reason for CW yielding less model degradation than PGD is likely due to its parameter being set to $\kappa = 0$. Empirically, however, CW has been shown to perform better at larger κ values, especially around $\kappa = 20$ [7]. Using this value, however, would make it more difficult to set up a fair comparison between CW and the other attacks. In addition, it would increase perturbation magnitude as well. Despite its higher computational cost, the early and drastic impact on both accuracy and TCRR highlights CW's strong performance.

The experiments demonstrate that not all adversarial attacks are created equal; each method offers a unique trade-off between effectiveness, efficiency, and transferability. FGSM, while computationally trivial, is relatively weak and inconsistent across models but has high attack transferability. PGD emerges as the most destructive, consistently reducing both accuracy and confidence across perturbation magnitudes with statistical significance. CW, although slightly weaker in degradation metrics than PGD, produces the most imperceptible adversarial examples. However, CW has low attack transferability, which confirms the precision of its optimization-based design for its source model.

An interesting observation is the pronounced difference in robustness between the two CNN architectures. ResNet-50 consistently displayed higher RA and TCRR under all attack types, especially against FGSM and PGD. This suggests that architectural features like residual connections may implicitly offer robustness benefits. However, even ResNet-50 ultimately failed under attacks with higher perturbation magnitudes, indicating that architectural resilience is not sufficient on its own.

In terms of transferability, FGSM displayed the strongest cross-model generalization. This likely stems from its simplistic gradient-based perturbation, which affects high-variance directions that are shared across models. CW's poor transferability highlights a key limitation of optimization-based attacks—they are highly tuned to the specific model and input. While this makes the attack suitable for evaluating model-specific vulnerabilities, it is less ideal for black-box scenarios or robustness benchmarking across architectures. Another reason behind the poor performance also had to do with the instantiated value of κ as well. The original CW paper showed that a value of $\kappa \approx 20$ drastically increased transferability rate [7] but as stated prior, this would make for an unfair comparison between attacks.

A significant practical constraint was the high computational cost of stronger attacks. PGD and CW are both iterative, and tuning their parameters required experimentation and substantial GPU resources. To accomodate for this, it was necessary to run attacks on multiple devices and leverage the DTU HPC clusters to make such evaluations feasible [13]. This raises broader questions about real-world applicability.

Finally, the included metrics, RA and TCRR proved effective at quantifying the degradation of both the binary correctness and confidence in the true class. TCRR, in particular, highlighted subtle yet critical effects of attacks: even when predictions remained correct, confidence in the true class often plummeted. This shows that accuracy alone is insufficient for evaluating adversarial robustness, especially in high-stakes applications where model confidence matters.

6.2 Societal and Ethical Reflections

The observed high transferability of simple adversarial attacks like FGSM raises critical concerns beyond academic curiosity. While such attacks are less imperceptible compared to more sophisticated methods like CW, their ease of use and effectiveness across different architectures make them particularly relevant in real-world adversarial scenarios. In our experiments, FGSM-crafted adversarial examples generated on VGG16 successfully degraded the performance of ResNet-50, which is a deeper and more complex model. This suggests that even relatively unsophisticated attacks can generalize across a non-trivial gap in model architecture and representational power.

This has direct implications for safety-critical systems such as autonomous vehicles and surveillance infrastructure. In these settings, robustness is not just a performance metric: it is a matter of public safety or civil rights. If adversarial examples crafted using open-source models can degrade the performance of proprietary or production-grade models deployed in self-driving cars, then the bar for launching a real-world black-box attack is alarmingly low. Similarly, in surveillance contexts, even crude perturbations applied to clothing, accessories, or makeup could help individuals evade detection by a range of face or object recognition systems, regardless of their internal architecture. The fact that high transferability exists between two such different CNNs in our experiments suggests a dangerous degree of cross-system vulnerability.

Furthermore, this underscores a key ethical tension in adversarial research. While publishing simple, transferable attacks contributes to scientific transparency and benchmarking, it also lowers the barrier for malicious use, especially since the required resources for crafting such attacks are minimal. FGSM can be implemented and executed in a matter of seconds, even on commodity hardware. In contrast, crafting imperceptible examples with CW requires fine-tuning and substantial compute, which naturally limits its accessibility.

Ultimately, these findings point to the importance of not overestimating the safety of current ML systems based solely on evaluations using imperceptible or high-cost attacks. The threat landscape includes low-cost, high-transferability methods that pose realistic risks in real-world deployments. In that sense, the simplicity of an attack should not be conflated with harmlessness. If anything, it makes it more likely to succeed in this regard. Using github as workflow, reproducibility of results is ensured [12].

6.3 Further Considerations and Scope Limitations

While this project focused on evaluating the impact and transferability of adversarial attacks across different CNN architectures, several adjacent techniques and areas of research fall outside our scope but remain highly relevant. One major omission is adversarial training, which remains one of the most prominent and effective defense mechanisms. By injecting adversarial examples into the training process, models can gain increased robustness to both white-box and black-box attacks. However, adversarial training introduces additional computational overhead and can reduce clean accuracy, depending on attack diversity and training dynamics. Including such a mechanism would have provided useful insight into whether the observed vulnerabilities (including those exposed via attack transferability) could be mitigated effectively.

It is also important to acknowledge that we exclusively evaluated two CNN-based architectures: VGG16 and ResNet-50. Although they differ in depth and structural design, they both fall under the same architectural paradigm. We did not include vision transformers or hybrid models, which are increasingly common in modern ML pipelines. These models may exhibit different forms of robustness or be susceptible to distinct failure modes under adversarial pressure. Broadening the architectural diversity of the study would help generalize the findings.

Moreover, while our work investigates transferability by testing white-box attacks on alternate models, we are fully aware that dedicated black-box attack techniques already exist, explicitly

developed for scenarios where model internals are not accessible. However, the project goal was not to replicate black-box attack literature, but to assess to what extent standard white-box attacks (FGSM, PGD, CW) generalize and transfer across architectures.

Other defense strategies, such as ensemble models or input transformation methods (e.g. JPEG compression, randomization or simply blurring the image) were also excluded from this study. While some of these may help reduce adversarial sensitivity, they often suffer from scalability limitations or domain-specific tuning requirements that make them difficult to deploy in practice. Including such techniques would have shifted the focus away from measuring raw vulnerability, which was the core aim of this project.

Finally, we did not address adaptive or iterative threat models where attackers refine their strategy over time based on model feedback. Our static evaluation setting does not capture the arms-race nature of adversarial machine learning as seen in deployed systems. Addressing this would require a different experimental design and was beyond the intended scope.

In summary, although our findings reveal clear and concerning vulnerabilities in common CNNs under simple white-box attacks, they represent just one slice of a much broader adversarial landscape. Future work should consider integrating defenses, alternative architectures, and more realistic attack scenarios to better approximate the complexities of real-world ML deployments.

7 Conclusion

This project evaluated the adversarialness of three attack methods (FGSM, PGD, and CW) as they were implemented on ResNet-50 and VGG16, and measured across multiple metrics, including RA, TCRR, and model transferability.

We found that PGD is the most powerful attack, rapidly degrading both accuracy and model confidence even at modest perturbation levels. CW creates the most efficient and imperceptible perturbations, though this comes at the cost of high computational demands and poor transferability. FGSM, in contrast, is the fastest and most transferable of the three, but also the least reliable in terms of adversarial degradation for model-specific targeting.

Among the two models, ResNet-50 consistently showed greater robustness than VGG-16 across all attack types. However, neither model could be considered truly robust when placed under sufficient adversarial pressure.

Finally, we found that TCRR is a valuable metric in its own right. It captures degradation in model certainty even when accuracy is preserved, offering a complementary and more nuanced perspective to traditional evaluation metrics.

Overall, the results emphasize the importance of thorough evaluation frameworks when studying adversarial attacks. Future work could include black-box attacks, targeted attacks, and defenses such as adversarial training. By exploring more nuanced metrics like TCRR, we can better understand, mitigate, and leverage the vulnerabilities of modern machine learning systems.

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9 Appendix

9.1 Robustness Ratio (RR) as a metric

While RR has gained popularity in recent years, there is no agreed standard definition of the metric within adversarial attacks literature. For instance, even recent papers like Zhou, et al., 2024 and Lyu, et al. 2024 define RR differently [14] [15]. This alone makes it difficult to find and choose an appropriate definition of the RR metric.

To see why the RR introduced in In Lyu, et al., 2024, was not included, let us observe the metric and how the paper defines it. Generally speaking, it measures the stability of model output probabilities after small perturbations. To avoid confusion, we will refer to this RR as the Predicted Class Robust Ratio (PCRR). In other words, PCRR quantifies the fraction of data where the model's confidence in its top prediction remain within a certain range under small perturbations. Consider the following defintion of a model's robustness φ_i^{rob} with weight matrix w_i over input x_i:

$$\varphi_i^{\text{rob}}(w, x_i) \triangleq \forall\, \delta \text{ s.t. } \|\delta\|_2 \leq \epsilon \implies \left|P(Y = \hat{y}_i \mid x_i + \delta, w) - P(Y = \hat{y}_i \mid x_i, w)\right| \leq b \quad (9)$$

Here, $||\delta||_2$ is the L_2 norm of the perturbation, $\delta \in \mathbb{R}^n$ is the perturbation vector, ϵ is perturbation threshold, and b is the bound on the continous outcome of the model, either normalized or given as a probability. This definition allows RR to measure the stability of a continous output within the observed perturbation range, even if label prediction remains unchanged. RR is defined as follows:

$$\text{PCRR} \triangleq \frac{1}{N} \sum_{i=1}^{N} \varphi_i^{\text{rob}}(w, x_i) \quad (10)$$

N is again the total number of samples contained in the dataset. Using RR, it is possible to assess the proportion of samples that satisfy local robustness under smaller perturbations, reflecting the model's general robustness. It is a tool that allows the quantification of changes in model confidence. A higher RR indicates that a model is more resilient to small perturbations for most samples.

While PCRR, as introduced by Lyu et al. (2024), measures how stable the model's confidence is in its top prediction under small perturbations, it does not consider whether that prediction is actually correct. As such, a model can exhibit a high PCRR even if it is confidently wrong. This is useful for targeted adversarial attacks where the objective is to cause a machine learning model to misclassify an input as a specific, adversary-chosen target class. However, this is not within the scope of this paper. In contrast, our proposed TCRR focuses specifically on the confidence in the *true class*, offering a more meaningful measure of robustness in terms of correctness. This makes TCRR more suitable for our evaluation, where we are interested in the model's ability to maintain confidence in the ground truth under adversarial conditions.

For the definition proposed by Zhou, et al., 2024, their robustness ratio is part of a broader adversarial attack metric framework introduced in the same paper. Since we do not include those metrics, looking at their RR alone is not informative.

9.2 Model finetuning epoch history

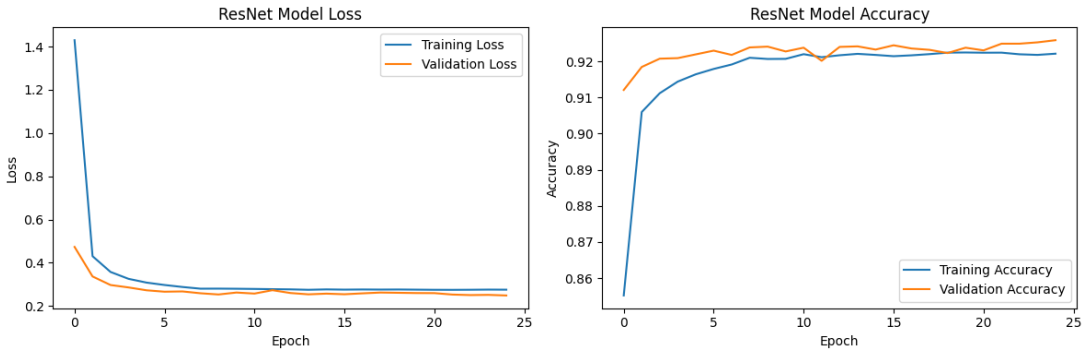


Figure 9: In the ResNet model loss and accuracy plots, a rapid decrease in training loss during the initial epochs can be observed, with the validation loss closely tracking it throughout training. Both curves stabilize at a low value (0.25), indicating efficient learning and minimal overfitting. The accuracy plot shows training and validation accuracy converging quickly, which is to be expected with finetuning / transfer-learning. The close alignment between the curves suggests that the model generalizes well to unseen data and has reached a stable performance early in training.

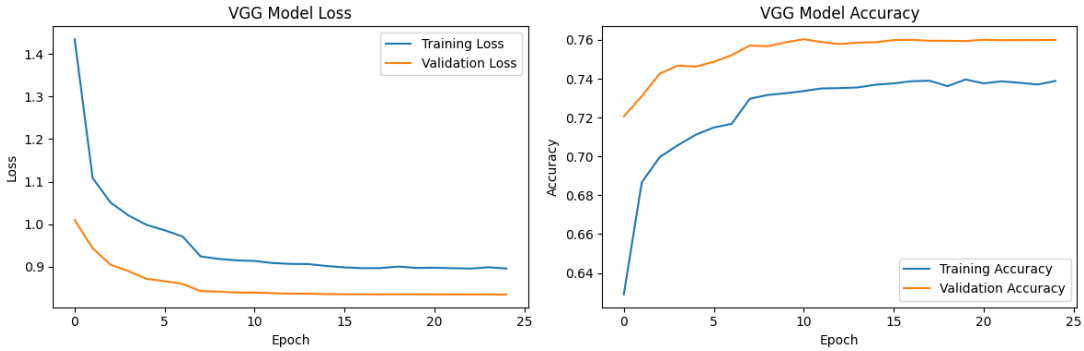


Figure 10: For the VGG model, the training loss decreases steadily but more slowly than in ResNet, and remains consistently higher than the validation loss. The validation loss plateaus early, suggesting that the model quickly stops learning, and struggles to fit the training data as effectively compared to ResNet-50. In the accuracy plot, the validation accuracy remains ahead of training accuracy by a small but persistent margin, leveling off around 76%. This pattern indicates potential underfitting, where the model does not fully capture the training data's complexity despite showing stable generalization.